

There are six main “disasters” due to rounding error that are well documented:

1. The Patriot and the Scud.
2. The short flight of the Ariane 5.
3. The Vancouver Stock Exchange.
4. Parliamentary elections in Schleswig-Holstein.
5. Knight Capital Group
6. The sinking of the Sleipner A offshore platform
7. What's 77.1 x 850? Don't ask Excel 2007

I just found #5 doing some search:

https://medium.com/@sohail_saifi/the-floating-point-error-that-sank-a-billion-dollar-hedge-fund-60fba9b54ae8

"August 1, 2012. Knight Capital Group lost \$440 million in 45 minutes. Not because of a market crash. Not because of a bad bet. Because of a floating point rounding error in their trading algorithm that had been lurking in their codebase for years."

This webpage is good and is the only listing #6

<https://diamhomes.ewi.tudelft.nl/~kvuik/wi211/disasters.html>

What's 77.1 x 850? Don't ask Excel 2007

<https://www.theregister.com/off-prem/2007/09/26/whats-771-x-850-dont-ask-excel-2007/286963>

Here is a copy paste from

<https://mathworld.wolfram.com/RoundoffError.html>

An egregious example of roundoff error is provided by a short-lived index devised at the Vancouver stock exchange (McCullough and Vinod 1999). At its inception in 1982, the index was given a value of 1000.000. After 22 months of recomputing the index and truncating to three decimal places at each change in market value, the index stood at 524.881, despite the fact that its "true" value should have been 1009.811.

Other sorts of roundoff error can also occur. A notorious example is the fate of the Ariane rocket launched on June 4, 1996 (European Space Agency 1996). In the 37th second of flight, the inertial reference system attempted to convert a 64-bit floating-point number to a 16-bit number, but instead triggered an overflow error which was interpreted by the guidance system as flight data, causing the rocket to veer off course and be destroyed.

The Patriot missile defense system used during the Gulf War was also rendered ineffective due to roundoff error (Skeel 1992, U.S. GAO 1992). The system used an integer timing register

which was incremented at intervals of 0.1 s. However, the integers were converted to decimal numbers by multiplying by the **binary** approximation of 0.1,

$$0.00011001100110011001100_2 = \frac{209\,715}{2\,097\,152}.$$

As a result, after 100 hours (3.6×10^6 ticks), an error of

$$\left(\frac{1}{10} - \frac{209\,715}{2\,097\,152}\right)(3600 \cdot 100 \cdot 10) = \frac{5625}{16\,384} \approx 0.3433 \text{ second}$$

had accumulated. This discrepancy caused the Patriot system to continuously recycle itself instead of targeting properly. As a result, an Iraqi Scud missile could not be targeted and was allowed to detonate on a barracks, killing 28 people.

Chartier, T. "Devastating Roundoff Error." *Math. Horizons* **13**, No. 4, 11, Apr. 2005. European Space Agency. "Ariane 501 Inquiry Board Report." Paris, 19 July 1996. <http://ravel.esrin.esa.it/docs/esa-x-1819eng.pdf>. McCullough, B. D. and Vinod, H. D. *J. Economic Lit.* **37**, pp. 633-665, June 1999. Quinn, K. "Even Had Problems Rounding Off Figures? This Stock Exchange Has." *The Wall Street Journal* **202**, No. 91, p. 37, Nov. 8, 1983. Skeel, R. "Roundoff Error and the Patriot Missile." *SIAM News* **25**, 11, Jul. 1992. *The Toronto Star*. November 19, 1983. United States General Accounting Office. "GAO/IMTEC-92-26 Patriot Missile Software Problem." 1992. <http://www.fas.org/spp/starwars/gao/im92026.htm>. Wilkinson, J. H. "Modern Error Analysis." *SIAM Rev.* **13**, 548-568, 1971. Wilkinson, J. H. *Rounding Errors in Algebraic Processes*. New York: Dover, 1994.

Copy pasted from <https://web.ma.utexas.edu/users/arbogast/misc/disasters.html>

1. The Patriot and the Scud.

Sources

1. General Accounting Office Report GAO/IMTEC-92-26.
2. Robert Skeel, "Roundoff Error Cripples Patriot Missile," *SIAM News*, July 1992.

On February 25, 1991, during the Gulf War, a Patriot missile defense system let a Scud get through. It hit a barracks, killing 28 people. The problem was in the differencing of floating point numbers obtained by converting and scaling an integer timing register. The GAO report has less than the full story. For that see Skeel's excellent article.

2. The short flight of the Ariane 5.

Source

1. Report by the Inquiry Board.

On June 4, 1996, the first Ariane 5 was launched. All went well for 36 seconds. Then the Ariane veered off course and self-destructed. The problem was in the Inertial Reference

System, which produced an operation exception trying to convert a 64-bit floating-point number to a 12-bit integer. It sent a diagnostic word to the On-Board Computer, which interpreted it as flight data. Finis.

Ironically, the computation was done by legacy software from the Ariane 4, and its results were not needed after lift-off.

3. The Vancouver Stock Exchange.

Sources

1. The Wall Street Journal November 8, 1983, p.37.
2. The Toronto Star, November 19, 1983.
3. B.D. McCullough and H.D. Vinod Journal of Economic Literature Vol XXXVII (June 1999), pp. 633-665. (References communicated by Valerie Fraysse)

In 1982 (I figure) the Vancouver Stock Exchange instituted a new index initialized to a value of 1000.000. The index was updated after each transaction. Twenty two months later it had fallen to 520. The cause was that the updated value was truncated rather than rounded. The rounded calculation gave a value of 1098.892.

4. Parliamentary elections in Schleswig-Holstein.

Source

1. [Rounding error changes Parliament makeup](#), Debora Weber-Wulff, The Risks Digest, Volume 13, Issue 37, 1992.

In German parliamentary elections, a party with less than 5.0% of the vote cannot be seated. The Greens appeared to have a cliff-hanging 5.0%, until it was discovered (after the results had been announced) that they really had only 4.97%. The printout was to two figures, and the actual percentage was rounded to 5.0%.

5. A few more examples

Try searching the [RISK Digest](#) with the key word "rounding" (David Goldberg's idea).

Here is a copy paste from

<https://catless.ncl.ac.uk/risks/13/37#subj4.1>

Rounding error changes Parliament makeup

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Tue, 7 Apr 1992 12:38:29 GMT

We experienced a shattering computer error during a German election this past Sunday (5 April). The elections to the parliament for the state of Schleswig-Holstein were affected.

German elections are quite complicated to calculate. First, there is the 5% clause: no party with less than 5% of the vote may be seated in parliament. All the votes for this party are lost. Seats are distributed by direct vote and by list. All persons winning a precinct vote (i.e. having more votes than any other candidate in the precinct) are seated. Then a complicated system (often D'Hondt, now they have newer systems) is invoked that seats persons from the party lists according to the proportion of the votes for each party. Often quite a number of extra seats (and office space and salaries) are necessary so that the seat distribution reflects the vote percentages each party got.

On Sunday the votes were being counted, and it looked like the Green party was hanging on by their teeth to a vote percentage of exactly 5%. This meant that the Social Democrats (SPD) could not have anyone from their list seated, which was most unfortunate, as the candidate for minister president was number one on the list, and the SPD won all precincts: no extra seats needed.

After midnight (and after the election results were published) someone discovered that the Greens actually only had 4,97% of the vote. The program that prints out the percentages only uses one place after the decimal, and had *rounded the count up* to 5%! This software had been used for *years*, and no one had thought to turn off the rounding at this very critical (and IMHO very undemocratic) region!

So 4,97% of the votes were thrown away, the seats were recalculated, the SPD got to seat one person from the list, and now have a one seat majority in the parliament. And the newspapers are clucking about the "computers" making such a mistake.

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There is also a PDF about the The Vancouver Stock Exchange
Sources: <https://www5.in.tum.de/~huckle/Vancouv.pdf>

This repeat more or less the same things:

<https://my.eng.utah.edu/~cfurse/ece6340/LECTURE/links/Embarrassments%20due%20to%20rounding%20error.htm>

A more intense read would be:

<https://www5.in.tum.de/~huckle/bugse.html>